COMPUTER VISION PROJECT

CIFAR-10 Image Classification with ResNet50 Transfer Learning

Master School Institute of Technology Berlin

STUDENT
Armin Kajevic

INSTRUCTOR
Lars Becker

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DATASET

CIFAR-10 image classification across ten object categories.

METHOD

Transfer learning with ResNet50 and controlled fine-tuning.

OUTCOME

64.42% final test accuracy with transparent error analysis.

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Automating Visual Recognition

The project addresses a classic computer vision problem: assigning small color images to predefined categories. CIFAR-10 is compact enough for controlled experimentation and challenging enough to reveal real model trade-offs.

10
OBJECT CLASSES

60k
BENCHMARK IMAGES

DECISION AREA	PROJECT CHOICE	MANAGEMENT RELEVANCE
Dataset	CIFAR-10, ten classes	Uses a recognized benchmark with measurable outcomes and comparable standards.
Model Strategy	Transfer learning with ResNet50	Reduces development effort by reusing pre-trained visual representations.
Execution Platform	Google Colab with GPU	Enables fast experimentation without investing in local training infrastructure.
Evaluation	Accuracy, loss, confusion matrix, examples	Supports both quantitative performance review and qualitative risk assessment.

Footnote [1]: CIFAR-10 originates from the University of Toronto dataset collection and is widely used for image classification experiments.

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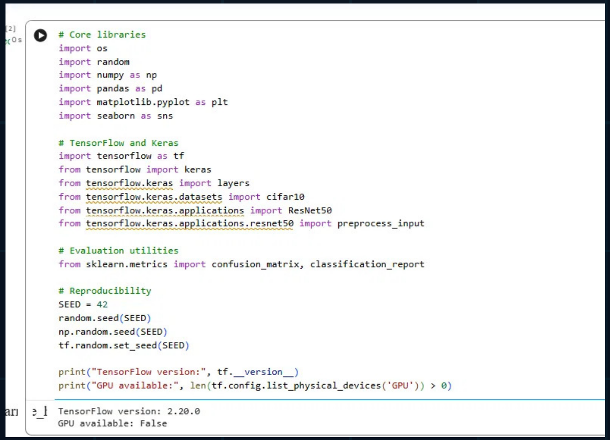
Complete Computer Vision Pipeline

The notebook was structured as a reproducible workflow, not as an isolated model training run. Each stage supports traceability from raw data to final business interpretation.

From dataset access to evidence-based evaluation.

STAGE	WHAT WAS DONE	WHY IT MATTERS
Data Loading	CIFAR-10 loaded through TensorFlow/Keras	Standardized and reproducible access.
Data Reduction	Training set limited to 10,000 images	Balances project scope and runtime efficiency.
Preprocessing	Inputs adapted for ResNet50	Ensures compatibility with the pre-trained model.
Transfer Learning	ResNet50 used with ImageNet weights	Leverages prior visual feature learning.
Evaluation	Metrics, confusion matrix, examples	Provides evidence beyond one headline score.

NOTEBOOK EVIDENCE



```python
# Core libraries
import os
import random
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# TensorFlow and Keras
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.datasets import cifar10
from tensorflow.keras.applications import ResNet50
from tensorflow.keras.applications.resnet50 import preprocess_input

# Evaluation utilities
from sklearn.metrics import confusion_matrix, classification_report

# Reproducibility
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
tf.random.set_seed(SEED)

print("TensorFlow version:", tf.__version__)
print("GPU available:", len(tf.config.list_physical_devices('GPU')) > 0)
```

```
TensorFlow version: 2.20.0
GPU available: False
```

The codebase used TensorFlow and Keras components for dataset access, model construction, training control, and evaluation.

Footnote [2][3]: ResNet50 and CIFAR-10 were accessed through the TensorFlow/Keras ecosystem and Keras Applications documentation.

Train Head First, Then Fine-Tune

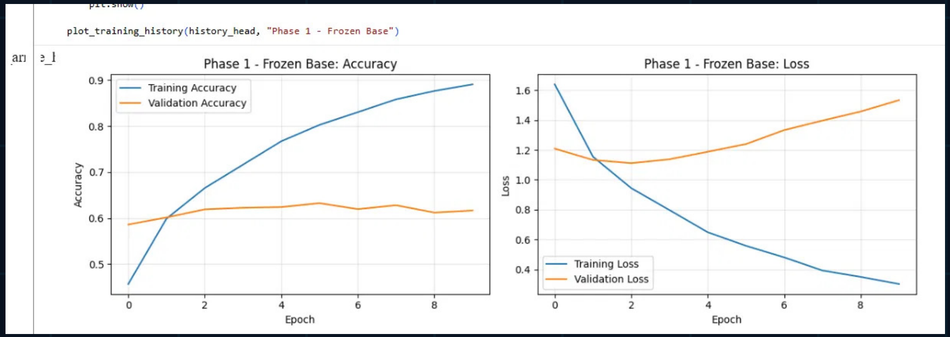
The model followed a controlled two-phase transfer learning strategy. This reduced early training risk, then increased flexibility once the custom classification head had learned CIFAR-10-specific boundaries.

PHASE 1	**Frozen ResNet50 base**	Only the custom classification head learned the ten CIFAR-10 categories.
PHASE 2	**Fine-tuned ResNet50**	The base model was unfrozen and adapted using a lower learning rate.
STABILITY	**BatchNormalization kept frozen**	This helped reduce training instability during fine-tuning.

Professional transfer learning is staged: first control, then adaptation.

Footnote [3][4]: ResNet50 transfer learning reused ImageNet feature representations through Keras Applications.

PHASE 1 TRAINING CURVE



The frozen-base phase quickly learned the new head, but validation behavior showed that fine-tuning was still needed for better generalization.

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Results Reveal Generalization Limits

The final model reaches a credible baseline for a constrained 10,000-image training subset, while the training-validation spread makes the overfitting signal visible and measurable.

FINAL MODEL OUTCOME

64.42%

Final test accuracy on CIFAR-10. This is the headline model result and the main decision metric for the baseline.

1.5049 FINAL TEST LOSS

EXECUTIVE READING

Transparent baseline, but not yet production-level accuracy.

ACCURACY COMPARISON

Test accuracy **64.42%**

Generalization result on unseen test data.

Training accuracy **98.6%**

Model learned the reduced training subset very strongly.

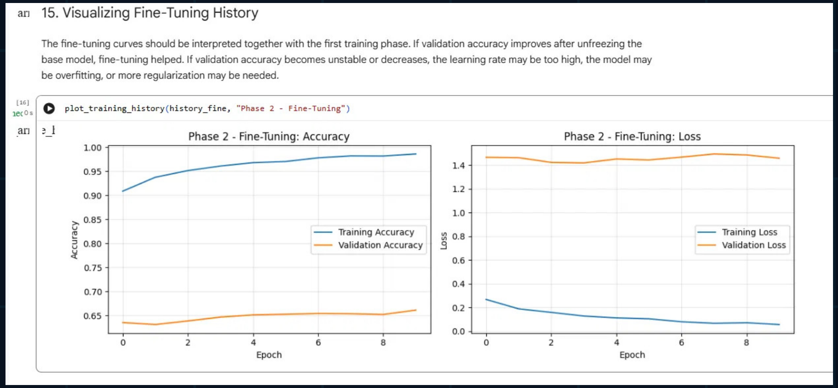
Validation accuracy **66.1%**

Validation stabilized close to the final test result.

32.5 pp

Approximate train-validation gap after fine-tuning.

FINE-TUNING EVIDENCE



The curve confirms the story behind the metrics: training performance keeps rising, while validation performance plateaus near the mid-60% range.

Best next action: stronger augmentation, regularization, and learning-rate control.

Footnote: Reported values come from the completed CIFAR-10 ResNet50 notebook: test accuracy 64.42%, test loss 1.5049, training accuracy about 98.6%, validation accuracy about 66.1%.

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Similar Classes Drive Most Mistakes

The confusion matrix and prediction examples show that the model is not failing randomly. Mistakes concentrate where CIFAR-10 images are small, low-resolution, and visually similar.

MODERN ERROR PATTERN VIEW

ANIMALS

Cat vs. dog

Texture, pose, and background overlap increase fine-grained recognition risk.

VEHICLES

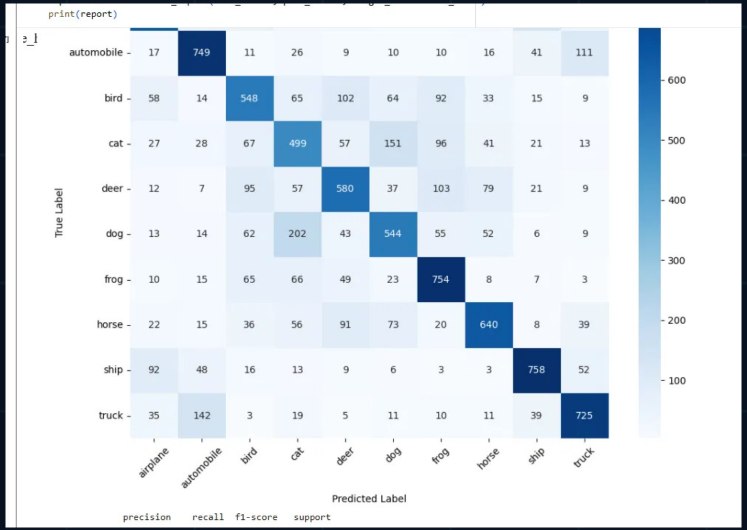
Auto vs. truck

Shape, viewpoint, and scale similarity make vehicle boundaries harder.

EXECUTIVE READING

Fix confusing pairs with stronger augmentation, regularization, and learning-rate control.

CONFUSION MATRIX EVIDENCE



The diagonal remains the dominant signal, while off-diagonal cells reveal where visually adjacent categories overlap.

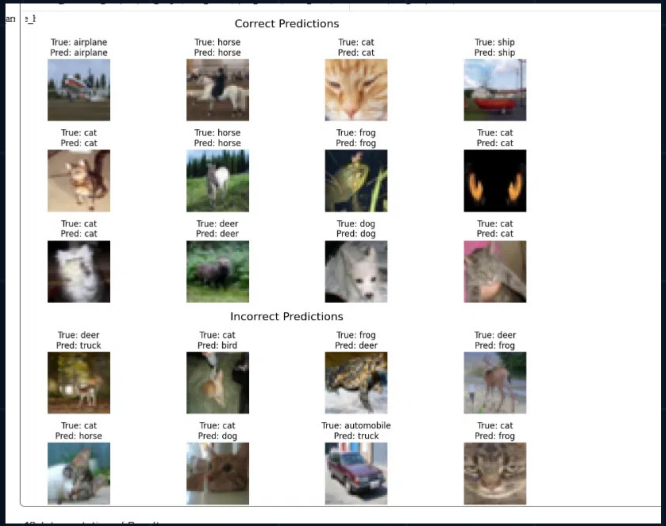
SIGNAL

Class-level risk map

USE

Prioritize fixes

PREDICTION EXAMPLES



Qualitative examples make the quantitative result explainable for non-technical stakeholders.

MEANING

Errors are interpretable

ACTION

Augment and tune

Footnote: Error findings are based on the confusion matrix and prediction examples generated in the completed project notebook.

Strong Baseline, Clear Next Steps

The project demonstrates a complete and reproducible computer vision workflow using transfer learning. The final model is a credible baseline with transparent limitations and actionable improvement paths.

64.42%

FINAL TEST ACCURACY WITH EXPLAINABLE ERROR BEHAVIOR

STRENGTH	EVIDENCE
Complete workflow	Data loading, preprocessing, training, fine-tuning, and evaluation.
Professional evaluation	Accuracy curves, test metrics, confusion matrix, and prediction examples.
Transparent limitations	Overfitting identified through training-validation behavior.
Future direction	Data augmentation, stronger regularization, and learning-rate tuning.

RECOMMENDED NEXT EXPERIMENT

Increase training diversity with augmentation, add stronger regularization, and compare learning-rate schedules before scaling to a larger model or full dataset run.

INSTITUTIONAL SIGN-OFF

Master School Institute of Technology Berlin

COURSE
Computer Vision

INSTRUCTOR
Lars Becker

PRESENTATION DATE
28.05.2026

STUDENT SIGNATURE

Armin Kajevic

References: [1] University of Toronto, The CIFAR-10 Dataset. [2] Keras, CIFAR-10 small images classification dataset. [3] TensorFlow/Keras, ResNet50 Documentation. [4] Keras Applications Documentation.